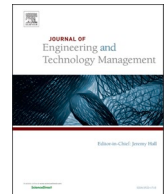




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## Examining the role of team boundary buffering as a proactive and reactive stress intervention for boundary spanning new product teams

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### ABSTRACT

In today's rapidly evolving business landscape, the capacity of new product development (NPD) teams to innovate hinges significantly on their ability to leverage external networks, known as 'team boundary spanning'. While this activity is traditionally viewed as beneficial, its potential to exacerbate job stress and consequently diminish team satisfaction and NPD performance demands closer examination. Our research extends this area of inquiry by exploring the impact of team boundary spanning on job stress within NPD teams. Drawing on the conservation of resources theory, we hypothesize that team boundary spanning increases job stress which, in turn, decreases job satisfaction and NPD performance. This study also investigates the role of team boundary buffering as both a proactive and reactive intervention to mitigate the stress experienced by teams. Data were collected from 140 NPD projects in high and medium-high technology firms. Results reveal that team boundary spanning indeed elevates job stress and that high levels of stress correlate with declines in product quality and job satisfaction. Additionally, our results indicate that while team boundary buffering is not an effective proactive stress intervention, it is beneficial as a reactive intervention to the stress precipitated by boundary spanning activities.

### 1. Introduction

Given the complex and interconnected nature of innovation today, team boundary spanning has become a common and indispensable activity in new product development (NPD) settings, making it essential for both team success and project outcomes (Zhang and Li, 2021; Kaiji et al., 2022). Boundary spanning involves proactive efforts by team members to reach beyond their immediate confines to secure essential resources, knowledge, and support crucial for innovation (Yan and Louis, 1999; Howell and Shea, 2006; Carbonell and Rodriguez-Escudero, 2019). By engaging with both internal and external groups, NPD teams secure vital resources, gather market insights, and stay informed about technological advancements and stakeholder expectations, ensuring that their projects align with external demands and organizational goals (Blindenbach-Driessen et al., 2010). These connections also help teams build relationships, negotiate support, and coordinate efforts across various parties, both within and outside the company (Marrone,

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2010). As collaboration and knowledge exchange remain central to NPD success (Papa et al., 2020; Zhang and Min, 2022), the ability to effectively span boundaries is not just beneficial but essential for the performance and success of NPD projects.

However, the benefits of boundary spanning come at a cost. Several studies have shown that individual-level boundary spanning can increase job stress due to the added workload and the complexity of managing external relations and expectations (e.g., Marrone et al., 2007; Liu et al., 2024; Zhu et al., 2020; Wang et al., 2019; Wu et al., 2020). Despite these findings, little is known about whether these effects also occur at the team level (Leicht-Deobald et al., 2023). Given that NPD teams frequently engage in boundary spanning activities (Carbonell and Rodríguez-Escudero, 2023), it is crucial to examine how team boundary spanning impacts job stress in this context. Understanding this relationship is especially important, as unmanaged job stress can lead to serious consequences such as burnout, which the World Health Organization has recognized as an occupational risk factor (World Health Organization, 2019).

Against this backdrop, the primary objectives of our study are to examine whether team boundary spanning contributes to job stress within NPD teams and, if so, explore how this effect can be mitigated. Drawing on the COR theory (Hobfoll, 1989; Hobfoll et al., 2018), we propose that while boundary spanning is crucial for acquiring resources and achieving innovative outcomes in NPD, it can also drain valuable team resources, leading to job stress and negatively impacting job satisfaction and NPD performance.

To mitigate the negative effects of team boundary spanning on job stress, we examine the role of team boundary buffering, a set of practices aimed at shielding the team from excessive external pressures and disruptions that might otherwise hinder performance (Faraj and Yan, 2009; Marrone et al., 2018). According to boundary management literature, buffering is essential for maintaining focus and preventing resource overload caused by external demands (Drach-Zahavy and Somech, 2010; Kislov, 2018; Leicht-Deobald et al., 2022). In this study, we hypothesize that buffering can both mitigate the stress induced by boundary spanning and reduce its negative impact on job satisfaction and NPD performance. By proactively protecting and replenishing critical team resources (e.g., attention, energy), boundary buffering can help lower the likelihood that boundary spanning will lead to stress. Additionally, after stress has occurred, buffering can play a reactive role, helping to alleviate its negative effects on team outcomes.

This study offers important theoretical and practical contributions. First, from a theoretical perspective, our research contributes to a deeper understanding of the consequences of team boundary spanning by examining its negative effects in the context of NPD teams. While prior research has predominantly highlighted the positive outcomes of boundary spanning (see meta-analysis by Leicht-Deobald et al., 2023), we explore the less-studied impact of boundary spanning on job stress. Second, our study applies the COR theory (Hobfoll, 1989) to a new context—NPD teams—where its use has been limited, thereby contributing to theory testing. We test how boundary spanning depletes key team resources—such as energy, attention, and time—leading to increased job stress and, in turn, decreased satisfaction and NPD performance. Finally, by elucidating the interaction between team boundary spanning and boundary buffering, this research provides nuanced insights into the balancing act required in NPD teams. This involves managing the dual demands of extending boundaries to gather external resources and strategically buffering those boundaries to protect the team from potential disruptions and stress. Such a dynamic interplay is crucial for maintaining high team satisfaction and delivering strong NPD performance.

From a managerial perspective, by exploring the nuanced interplay between team boundary spanning, job stress, and boundary buffering, our study delivers actionable insights that can help NPD teams engaged in boundary spanning activities not only enhance their performance but also safeguard the mental health and satisfaction of their members.

Methodologically, our investigation employs a robust empirical approach, analyzing data from 140 NPD projects from high- and medium-high technology manufacturing firms. The firms for this study were selected from the Orbis Europe directory, a comprehensive database managed by Moody's Analytics. To test our research model, we use partial least squares-structural equation modeling (PLS-SEM), ensuring a precise evaluation of the interactions and effects within our theoretical framework.

In the following sections, we begin by reviewing the theoretical foundations of the study, including a discussion of the literature on team boundary spanning and boundary buffering, as well as the presentation of the COR theory, which frames our hypotheses. Next, we present the study's hypotheses, followed by a description of the methodology, data analysis, and results. Finally, we discuss the findings and outline their implications for both theory and practice, concluding the article with a description of the study's limitations and directions for future research.

## 2. Theoretical framework

### 2.1. Team boundary spanning and boundary buffering

Boundary management is crucial for the success of NPD teams, where balancing internal processes with external engagement is key to fostering innovation (Marrone, 2010). Boundary management can be understood as the processes and activities through which a team engages to define and manage its boundaries. This includes interactions with external parties as well as the activities that establish and define these borders (Dey and Ganesh, 2017). Our research focuses on two main components of boundary management: team boundary spanning and team boundary buffering, each playing a unique yet interconnected role in the team's external engagements and internal operations.

Team boundary spanning involves proactive efforts by team members to reach beyond their immediate confines to secure essential resources, knowledge, and support crucial for innovation. It encompasses several key activities including ambassador, scouting, and task coordination activities (Marrone, 2010). Ambassador activities involve promoting the NPD project and garnering support by establishing connections with external parties. Scouting activities are focused on acquiring project-related information, such as stakeholders' expectations, market trends, and technological opportunities. Task coordination involves negotiating resources, solving problems collaboratively, and harmonizing NPD activities with both internal and external groups (Ancona et al., 2009; Marrone,

2010). While significant research highlights the benefits of boundary spanning in terms of innovation and performance enhancements (Ancona and Caldwell, 1992), boundary spanning can also generate risks such as increased job stress and role conflict (Gibson and Dibble, 2013). For example, Faraj and Yan (2009) note that boundary spanning can lead to an influx of external requests, complicating team dynamics and increasing job stress. This view is supported by Wu et al. (2020), who argue that excessive external interference can undermine team performance. Despite these insights, research on the negative consequences of boundary spanning remains limited, with much focus on individual-level outcomes (Marrone et al., 2007; Wang et al., 2019).

Conversely, boundary buffering refers to practices aimed at protecting the team from potential disruptions and external pressures that might impede their focus on performance. Unlike boundary spanning, which involves active engagement with external parties, boundary buffering is a strategy of disengagement (Faraj and Yan, 2009). Buffering activities can be employed either in response to or in anticipation of disruptive forces in a team's external environment and include setting limits on information flow, interactions with external entities, and shielding the team from troubling information and political pressures (Ancona and Caldwell, 1988). These activities also involve monitoring requests made by external parties and determining how the team will respond to those demands (Ancona and Caldwell, 1988). Several studies emphasize the importance of buffering in maintaining team focus and preventing overload, which can result from excessive external demands (Marrone et al., 2018).

This refined understanding of boundary management forms the foundation of our study, which investigates how interactions between boundary spanning and boundary buffering influence job stress, team satisfaction, and NPD performance. In the dynamic context of NPD, both boundary spanning and buffering are essential (Leicht-Deobald et al., 2023). Boundary spanning allows teams to proactively secure essential external resources and information, critical for innovation amidst high uncertainty and rapid market changes (Carbonell and Rodríguez-Escudero, 2019 and 2023). Concurrently, boundary buffering safeguards the team from potential information overload and external disruptions (Faraj and Yan, 2009). These strategies together empower NPD teams to effectively balance external collaboration with internal stability, thereby enhancing their ability to handle external pressures. This equilibrium is vital for reducing job stress, increasing job satisfaction, and boosting new product performance, all of which are key to successful product development.

Our research addresses a significant gap identified in the literature (Leicht-Deobald et al., 2022) by exploring how the interplay between boundary spanning and buffering impacts job stress, job satisfaction, and NPD performance. Previous studies have often viewed these two activities as conflicting, with boundary spanning aimed at loosening boundaries and buffering designed to strengthen them (Dey and Ganesh, 2017; Drach-Zahavy and Somech, 2010). Contrary to the traditional view that teams must choose between these strategies, our study proposes that boundary spanning and buffering can synergistically enhance team well-being and performance. Drawing on the COR theory (Hobfoll, 1989), we hypothesize that although boundary spanning may initially deplete resources and increase job stress in NPD teams, implementing buffering activities can proactively safeguard or replenish these resources. This, in turn, reduces the likelihood that boundary spanning will lead to stress. Furthermore, following the experience of stress, boundary buffering can mitigate its negative effects on job satisfaction and NPD performance. The next section introduces the COR theory and details its relevance to our research context.

## 2.2. Conservation of resources theory

Developed by Hobfoll (1989), the COR theory posits that individuals and groups strive to acquire, maintain, and safeguard resources that contribute to goal achievement. Resources can include physical, psychological, social, or situational factors that facilitate goal attainment, reduce demands, and stimulate personal growth. Hobfoll (2001) categorizes resources as either instrumental (serving as means to achieve other resources or positive outcomes) or intrinsic (intrinsically valued as ends in themselves) (Singh et al., 2018).

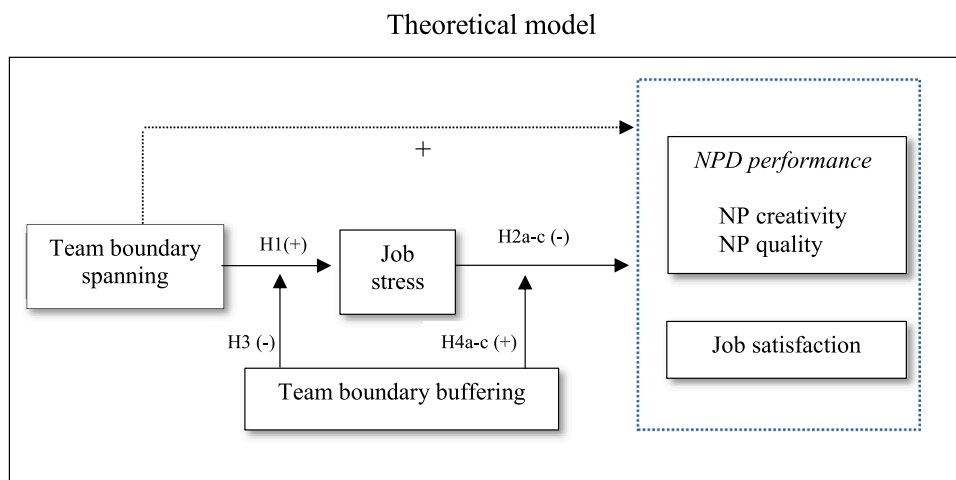


Fig. 1. Theoretical model.

According to the COR theory, stress arises when resources are threatened or depleted (Hobfoll, 1989). Drawing upon the COR theory, we perceive team boundary spanning as a stress-inducing factor. Team boundary spanning is recognized as a demanding and challenging activity that necessitates significant effort and time (Marrone et al., 2007). Consequently, these activities are likely to deplete a team's resources such as time, effort, and energy. Thus, based on the COR theory, team boundary spanning represents a threat to a team's valuable resources and can be regarded as a source of stress (see Fig. 1). In this study, job stress is defined as an unfavourable and undesirable state of mind (such as hopelessness, tension, or frustration) that frequently emerges when there is a disparity between the demands imposed on individuals within their work environment and their capacity to fulfil those demands (Le Fevre et al., 2003).

According to the COR theory, employees experiencing stress tend to conserve their remaining resources by reducing morale, diminishing commitment to the organization, and lowering performance efforts (Hobfoll, 1989). Building on this concept, we anticipate that job stress will negatively impact NPD performance (measured in terms of new product quality and new product creativity), and job satisfaction. New product quality refers to how effectively the new product meets the needs of target customers (Koufteros et al., 2005; Sheremata, 2000). New product creativity is defined as the extent to which a new product is innovative and has the potential to influence thinking and practices (Moorman and Miner, 1997). Job satisfaction is defined as "the team's shared attitude toward its task and its work environment" (Mason and Griffin, 2002, p. 1).

As shown in Fig. 1, team boundary buffering is expected to mitigate the positive relationship between team boundary spanning and job stress. Team boundary buffering activities allow NPD teams to work with minimal distraction from external parties and to refuse external requests when the latter are not perceived as legitimate (Ancona and Caldwell, 1988; Kislow, 2018), thus helping teams protect valued resources (e.g., attention, energy). Therefore, from a COR perspective, it is reasonable to argue that the feeling of stress created by team boundary spanning may be reduced by engaging in team boundary buffering activities.

Lastly, COR theory proposes that people who have more resources can deal with stress better (Hobfoll, 1989). Drawing on this notion, we expect team boundary buffering to reduce the negative impact of job stress on job satisfaction and NPD performance. Thus, it has been documented that boundary buffering activities can increase a team's psychological safety (Faraj and Yan, 2009) and vigor (Leicht-Deobald et al., 2022), thus providing boundary spanning NPD teams with additional valuable resources to adapt and react positively to felt job stress.

### 3. Hypotheses development

#### 3.1. Effect of team boundary spanning on job stress

Despite the numerous advantages, team boundary spanning can contribute to stress for several reasons. Firstly, boundary spanning activities require significant effort and time (Gibson and Dibble, 2013), which can pose challenges to the team's functioning. This aligns with the findings of Marrone et al. (2007), who suggest that the benefits of increased boundary spanning come at the expense of experiencing role overload. Moreover, boundary spanning roles have been associated with challenges such as role stress, conflict, and ambiguity (Katz and Khan, 1978). NPD teams engaged in boundary spanning activities interact with various external parties, including management, clients, and other departments, each with different and often conflicting demands and expectations. Striving to meet these divergent expectations can consume a significant portion of the team's resources (such as time and energy) and create a sense of ambiguity and anxiety within the team (Gibson and Dibble, 2013). Finally, engaging in team boundary spanning has the potential to create an overload of external requests and interferences (Zhang et al., 2022), reducing the available resources that team members can mobilize to do other necessary tasks (Gibson and Dibble, 2013). Thus, in keeping with the COR theory, we posit that:

**H1.** . There is a positive relationship between team boundary spanning and job stress.

#### 3.2. Effect of job stress on NPD performance and job satisfaction

According to the COR, when employees are stressed, they may resort to preserving their remaining resources by lowering their morale, weakening their loyalty to the organization, and minimizing their performance efforts (Hobfoll, 1989; Hobfoll et al., 2018). In keeping with this idea, it has been posited that stressed employees are likely to reduce investment in creative activities and opt for relaxation in order to restore resources (de Jonge et al., 2012) which suggests a negative effect of job stress on new product creativity. This effect is in fact evidenced in the work of Amabile et al. (2002) who found that intensive workloads and time pressures led to professional workers being half as creative as they would otherwise be. Other scholars have also found a negative effect of role stress on employee creativity (Çekmecelioglu and Günsel, 2011; Zhu et al., 2020). Additionally, because of their motivation to preserve their resource reservoir, stressed employees are more likely to shy away from tasks that require time and effort (Jehn, 1995) and to show a lower willingness to deliver quality work (Humborstad et al., 2007) all of which is likely to result in lower new product quality. Indeed, in the field of NPD, job stress has been associated with lower quality work and more mistakes (Barczak and Wilemon, 2001). Finally, because of their low morale, it may be difficult for stressed employees to achieve a feeling of personal accomplishment and growth in their jobs (Barczak and Wilemon, 2001) and thus, to feel satisfied at work. Furthermore, anxious employees are less willing to assist other team members, resulting in an extremely unpleasant and unsatisfactory team experience (Humborstad et al., 2007). In the area of NPD project teams, job stress has been found to have a negative effect on team member satisfaction (Barczak and Wilemon, 2003). Based on the previous discussion, we propose that:

**H2.** . There is a negative relationship between job stress and NPD performance and job satisfaction such that higher levels of stress will decrease a) new product creativity, b) new product quality and c) job satisfaction.

### 3.3. Moderating effects of team boundary buffering

In this study, we propose that engaging in boundary buffering activities can mitigate the positive influence of team boundary spanning on job stress. Team boundary buffering involves regulating the flow of information and requests from external parties into the group (Ancona and Caldwell, 1988). This entails making decisions regarding the sources from which the group will accept input, the amount of input that will be considered, the format of the input, and when the inflow of input should cease (Ancona and Caldwell, 1988, p. 477). This can be particularly helpful for boundary spanning teams for whom, as mentioned earlier, workload and other stressors are often very high (Leicht-Deobald et al., 2022). Boundary buffering activities also involve keeping troubling information and political pressures away from the teams so that team members do their work with minimal distraction (de Vries et al., 2022). Lastly, team buffering involves disregarding external requests that are deemed illegitimate, ensuring that team members are not excessively burdened while striving to achieve team objectives. By possessing these characteristics, boundary buffering enables boundary spanning NPD teams to eliminate, deflect, or minimize irrelevant obstacles, pressures, and issues that are unrelated to their tasks, thereby safeguarding and preserving valuable team resources. Consequently, it is reasonable to anticipate that proactive engagement in team boundary buffering will reduce the likelihood of boundary spanning NPD teams experiencing job stress. Therefore, we propose the following hypothesis:

**H3.** : Team boundary buffering negatively moderates the relationship between team boundary spanning and job stress. Specifically, when boundary buffering is high, the positive influence of team boundary spanning on job stress will be weaker compared to when boundary buffering is low.

After job stress is recognized, a decision to engage in boundary buffering activities can help reduce the deleterious effects of job stress on NPD team performance. This premise is based on prior research which suggests that engaging in team boundary buffering can bring new valuable resources into a team (Faraj and Yan, 2009; Leicht-Deobald et al., 2022) thus increasing the pool of resources that teams have to cope with the negative effects of stress. In this respect, it has been contended that boundary buffering activities create a protected internal atmosphere for the team members, thereby contributing to psychological safety (Faraj and Yan, 2009). Psychological safety has been recognized as a valuable defense mechanism against the draining impact of job stress (Mansour et al., 2022). Thus, it is contended that teams with higher levels of psychological safety tend to foster a more positive emotional climate (Afsar et al., 2021). In such teams, members can find emotional support, gain confidence, and muster the courage to confront challenges more effectively. Also, in psychologically safe environments, team members do not have to worry or be afraid of being punished for saying or doing the wrong thing which, according to Zhou and Chen (2021) can consume significant personal resources (such as time and energy) and eventually lead to emotional exhaustion. Also, team boundary buffering activities can generate a greater level of team vigor. Hence, Leicht-Deobald et al. (2022) discovered that teams practicing boundary buffering exhibit elevated levels of enthusiasm, liveliness, and excitement when engaging in team tasks. As noted by Hsieh and Huang (2022), vigor, as an energy resource, can enhance employees' engagement with their jobs and work environment. This increased engagement empowers employees to feel a greater sense of control over their work and fosters more positive attitudes towards their job. From a COR perspective, we argue that the additional new resources brought in by boundary buffering activities will help NPD teams better adapt and cope with felt job stress, thus reducing the negative effects of stress on job satisfaction and NPD performance. Thus, we propose that:

**H4.** : Team boundary buffering positively moderates the relationship between job stress and job satisfaction and NPD performance such that job stress will have a weaker negative influence on (a) new product quality, (b) new product creativity and (c) job satisfaction when boundary buffering is high than when it is low.

## 4. Methodology

### 4.1. Sample and data collection

In defining the sampling frame, we focused on industrial sectors classified as high technology and medium-high technology by EUROSTAT. Although the food and beverage manufacturing sector is classified as low technology by EUROSTAT, it was included in the sampling frame due to its high R&D spending (MAPA Spanish Ministry of Agriculture, Fisheries and Food, 2021). To capture the full spectrum of team boundary spanning activities, we excluded firms with fewer than 35 employees. This decision was based on the premise that smaller firms have limited resources and organizational capacities, which constrain their ability to fully engage in extensive boundary spanning activities (Baumann and Kritikos, 2016; van Oostrom and Fernández-Esquinas, 2017). Following the approach used in other innovation studies (Pérez-Luño et al., 2019; Ritala and Hurlmelinna-Laukkanen, 2013), we relied on the Orbis Europe directory (formerly known as Amadeus), a comprehensive database managed by Moody's Analytics,<sup>1</sup> to identify companies in our target industries. In total, there were 3796 firms with 35 or more employees in the target industries. We randomly selected 25 percent of the firms from each industry group, resulting in a final sample of 946 manufacturing firms.

To mitigate the potential bias caused by common method variance (Podsakoff et al., 2012), a two-stage data collection approach was employed. In the first stage, a questionnaire focusing on the boundary management activities of NPD teams, specifically team boundary spanning and boundary buffering, was sent by mail to the individuals responsible for NPD activities in each company within

<sup>1</sup> <https://www.moody.com/web/en/us/capabilities/company-reference-data/orbis.html>.

our sample. Respondents were asked to select a new product introduced to the market within the past three years and had been available for at least six months. They were also informed that a follow-up questionnaire would be sent as part of the same research project. A total of 146 completed surveys were obtained in the first round. In the second stage, a second survey was distributed via post and email to the 146 firms that had responded to the initial survey, collecting data on job stress and NPD team performance. Out of the 146 firms, 140 completed the second survey. The time interval between the two surveys varied from one to four months. Of the key informants responding to the survey, 45.2 % were R&D managers, 34.1 % were technical managers, 13.3 % were general managers and 7.4 % were managers of other departments.

Table 1 presents both the population and the sample sizes for each industry group. To assess potential response bias by industry, we conducted a two-proportion test to evaluate the differences between the sample's and the population's proportions for each industry sector. Significant disparities were observed in the 'computer, electronic and electrical manufacturing' sectors (NAICS codes 334 and 335). Notably, companies from these high-technology sectors are disproportionately represented in the sample, which aligns with expectations due to their technological intensity. Furthermore, to assess non-response bias for our two surveys, we analysed the differences between the early respondents (first third) and those who responded later (last third). Our analysis showed no significant variations in the average scores of the primary constructs of this study.

#### 4.2. Measurement variables

We utilized Carbonell and Rodríguez-Escudero's (2019) scale to measure team boundary spanning. This scale considers team boundary spanning as a second-order formative construct comprising seven reflective dimensions. The measurement items for these dimensions can be found in Table 2. The reflective dimensions of the team boundary scale exhibited good composite reliability (CR) and average variance extracted (AVE) values, surpassing the recommended thresholds of 0.70 and 0.50, respectively. To statistically confirm the formative nature of the team boundary spanning construct, we assessed multicollinearity among its dimensions using the variance-inflation factor (VIF) and ensured that the outer weights of all dimensions were significant at a p-value of less than 0.01.

To measure boundary buffering we employed three items adapted from Faraj and Yan's (2009) team buffering scale. Job stress was measured using five items from Dayan and Di Benedetto's (2011) job stress scale. New product creativity was assessed using five items from Rindfleisch and Moorman's (2001) new product creativity scale, while new product quality was measured using a three-item scale derived from Sarin and Mahajan (2001). Job satisfaction was operationalized with three items borrowed from Hackman and Oldham's (1980) growth satisfaction scale plus a fourth item that assessed the extent to which team members were satisfied with the way they had worked as a team. The inclusion of this fourth item is supported by research in NPD which highlights the importance of understanding team dynamics due to their significant impact on overall job satisfaction (Gelbard and Carmeli, 2009). Preliminary feedback and pilot testing of our modified scale confirmed the relevance of this item to measuring job satisfaction, thereby affirming the construct validity of the scale (Table 3).

All variables exhibited CR and AVE values that exceeded the recommended thresholds of 0.70 and 0.50, respectively. To assess discriminant validity, we verified that the square root of the AVE for each construct was greater than the corresponding correlations with any other construct. All possible pairs of constructs passed this test (Table 4).

The study incorporates two control variables namely, firm size and team size. Larger organizations typically possess ample R&D, marketing, and financial resources, which facilitate the successful development and commercialization of new products and services (Ali et al., 1995). Thus, we anticipate that firm size will positively impact new product creativity, new product quality, and job satisfaction. Firm size is measured based on the number of employees in the company. Additionally, the study controls for the influence of team size on NPD team performance, as prior research suggests that team size can positively affect new product quality (Weiss et al., 2013), new product creativity (Dayan and Colak, 2008), and job satisfaction (Panko and Kinney, 1998). The team size variable is measured by considering the number of individuals fully involved in the NPD team. Lastly, the theoretical model includes direct links between team boundary spanning and our three measures of NPD team performance. Thus, it has been argued that connecting with individuals outside the NPD team can stimulate new ideas and lead to higher levels of new product creativity (Björk, 2012; Hemphälä and Magnusson, 2012; Somech and Khalaili, 2014). Additionally, boundary spanning activities can help NPD teams gain essential input and feedback on project related matters, improving new product quality (Menon et al., 1997). Finally, while this study highlights potential negative outcomes associated with team boundary spanning, such as job stress, engaging in team boundary spanning can also

**Table 1**  
Population and sample distribution by industry: proportion test.

| NAICS codes   | Industrial sector                                                                      | Population |            | Sample |            |
|---------------|----------------------------------------------------------------------------------------|------------|------------|--------|------------|
|               |                                                                                        | N          | % of total | N      | % of total |
| 311, 312      | Food and beverages manufacturing                                                       | 203        | 21.5 %     | 23     | 16.4 %     |
| 325, 326      | Chemical and plastics product manufacturing                                            | 213        | 22.5 %     | 36     | 25.7 %     |
| 333           | Machinery manufacturing                                                                | 122        | 12.9 %     | 15     | 10.7 %     |
| 327, 331, 332 | Non-metallic mineral product, primary metal and fabricated metal product manufacturing | 195        | 20.6 %     | 21     | 15.0 %     |
| 334, 335      | Computer, electronic, and electrical manufacturing                                     | 109        | 11.5 %*    | 28     | 20.0 %*    |
| 336           | Transportation equipment manufacturing                                                 | 104        | 11.0 %     | 17     | 12.1 %     |
|               | TOTAL                                                                                  | 946        | 100 %      | 140    | 100 %      |

\* Significant differences:  $p < 0.05$

**Table 2**  
Team boundary spanning dimensions.

|                                                                                                                          | Mean (S.D.) |
|--------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Factor 1. Intra-firm ambassador</b>                                                                                   |             |
| Project leader or team members talked up the NPD project to individuals or groups in the company.                        | 5.59 (1.31) |
| Persuaded them that the NPD project was important.                                                                       | 5.63 (1.24) |
| Persuaded them to support the team's activities and decisions.                                                           | 5.30 (1.24) |
| Kept them informed of the progress on the team's activities to gain support for the project.                             | 5.54 (1.17) |
| Built a relationship with these groups.                                                                                  | 5.21 (1.21) |
| <b>Factor 2. Extra-firm ambassador</b>                                                                                   |             |
| Project leader or team members talked up the NPD project to individuals or groups outside the company.                   | 5.01 (1.69) |
| Persuaded them that the NPD project was important.                                                                       | 4.85 (1.76) |
| Persuaded them to support the team's activities and decisions.                                                           | 4.34 (1.84) |
| Kept them informed of the progress on the team's activities to gain support for the project.                             | 4.27 (1.62) |
| Built a relationship with these groups                                                                                   | 4.99 (1.69) |
| <b>Factor 3. Intra-firm scouting–1</b>                                                                                   |             |
| Found out whether individuals or groups inside the company supported the team's activities                               | 4.97 (1.44) |
| Found out their expectations about the new product project                                                               | 5.26 (1.29) |
| <b>Factor 4. Intra-firm scouting–2</b>                                                                                   |             |
| Asked them about new or emerging marketing information and trends.                                                       | 4.85 (1.54) |
| Asked them about new or emerging technical information and trends.                                                       | 4.72 (1.70) |
| Collected their perceptions about team and project progress <sup>a</sup>                                                 |             |
| Asked them about information regarding the company's strategy and climate that might impact the NPD project <sup>a</sup> |             |
| <b>Factor 5. Extra-firm scouting</b>                                                                                     |             |
| Found out whether individuals or groups outside the company supported the team's activities.                             | 4.35 (1.82) |
| Found out their expectations about the new product project.                                                              | 4.64 (1.87) |
| Asked them about new or emerging marketing information and trends.                                                       | 4.69 (1.88) |
| Asked them about new or emerging technical information and trends.                                                       | 4.84 (1.71) |
| Collected their perceptions about team and project progress.                                                             | 4.24 (1.88) |
| Asked them about information regarding changes or early signs of trouble in the external environment                     | 4.93 (1.79) |
| <b>Factor 6. Intra-firm task coordination</b>                                                                            |             |
| Negotiated resources (time, money, people) for the team with groups inside the company <sup>a</sup>                      |             |
| Coordinated with them development and commercialization activities with other individuals or departments.                | 5.20 (1.48) |
| Resolved with them technical and marketing problems that arose during the development process.                           | 5.76 (1.27) |
| Reviewed with them product concepts, preliminary product designs and ideas for marketing plans.                          | 5.49 (1.48) |
| <b>Factor 7. Extra-firm task coordination</b>                                                                            |             |
| Procured knowledge and skills relevant to the NPD project from groups outside the company.                               | 3.89 (1.92) |
| Coordinated with them development and commercialization activities with interested parties.                              | 4.00 (1.75) |
| Resolved with them technical and marketing problems that arose during the development process.                           | 4.47 (1.72) |
| Reviewed with them product concepts, preliminary product designs and ideas for marketing plans.                          | 4.30 (1.85) |

yield positive outcomes for the team, including increased influence, reputation, and the ability to make valuable contributions (Marrone, 2010) which, in turn, can enhance job satisfaction (Wang et al., 2019).

## 5. Analysis and results

### 5.1. Hypotheses testing

We used PLS-SEM for our statistical analysis (Smart PLS 4.0.8 software) (Ringle et al., 2022). PLS-SEM was selected primarily for its robustness in handling models that include both first-order and second-order constructs, as well as its capacity to accommodate both formative and reflective measurement models (Chin and Newsted, 1999). This flexibility is crucial for our study, which incorporates a mix of these constructs. Moreover, unlike covariance-based SEM, PLS-SEM does not require normally distributed data and is less sensitive to sample size limitations (Hair et al., 2019), a significant advantage given our sample size of only 140 NPD projects. Another reason for choosing PLS-SEM is its efficacy in managing complex models without imposing restrictions on measurement scales (Falk and Miller, 1992; Tenenhaus et al., 2005), which is vital given the extensive number of items used to measure team boundary spanning in this study. For hypothesis testing, bootstrapping (5000 subsamples) was used to generate the standard errors and t statistics. Table 5 shows the path coefficients ( $\beta$ ) for the two estimated models and the variance explained ( $R^2$ ) in the dependent variables.

Model 1 allows us to test the hypotheses regarding the main effects. Our results show that team boundary spanning positively influences job stress ( $\beta = 0.20$ ,  $p < 0.05$ ), thereby supporting H1. Additionally, job stress has a significant negative effect on both new product quality ( $\beta = -0.35$ ,  $p < 0.05$ ) and job satisfaction ( $\beta = -0.33$ ,  $p < 0.05$ ), thus supporting H2b and H2c, respectively. However, job stress has no influence on new product creativity ( $\beta = -0.03$ ). Therefore, H2a is not confirmed.

Model 2 examines the moderating effects of team boundary buffering. The impact of team boundary buffering emerges as not significant in the relationship between team boundary spanning and job stress ( $\beta = -0.06$ ), such that we cannot support H3. However, we do find support for H4a, H4b and H4c. Team boundary buffering significantly moderates the relationships between team boundary spanning and new product creativity ( $\beta = 0.30$ ,  $p < 0.05$ ), new product quality ( $\beta = 0.25$ ,  $p = 0.07$ ) and job satisfaction ( $\beta = 0.20$ ,

**Table 3**

Measurement scales.

|                                                                                                             | Mean (S.D.)    |
|-------------------------------------------------------------------------------------------------------------|----------------|
| <b>Team boundary buffering</b>                                                                              |                |
| Team leader or project members absorbed outside pressures so that the team could work free of interference. | 5.17 (1.50)    |
| Prevented outsiders from overloading the team with too many requests or unproductive information.           | 5.02 (1.39)    |
| Helped team members to manage the demands placed on them by other organizational units.                     | 4.98 (1.50)    |
| <b>Job stress</b>                                                                                           |                |
| Team members felt overwhelmed by work during this project.                                                  | 4.23 (1.56)    |
| Team members experienced tension during this project.                                                       | 4.73 (1.51)    |
| Team members felt that things were out of control during this project.                                      | 3.10 (1.54)    |
| Team members sometimes felt like giving up on the job during this project.                                  | 1.96 (1.13)    |
| Team members felt pressured while working on this project.                                                  | 4.18 (1.68)    |
| <b>New product creativity</b>                                                                               |                |
| The product offers new ideas to the category.                                                               | 5.64 (1.18)    |
| It is very creative.                                                                                        | 5.01 (1.49)    |
| It is very interesting.                                                                                     | 5.73 (1.10)    |
| It is capable of generating ideas for other products.                                                       | 5.56 (1.30)    |
| It promotes fresh thinking.                                                                                 | 5.22 (1.35)    |
| <b>New product quality</b>                                                                                  |                |
| The product is reliable.                                                                                    | 6.10 (0.84)    |
| The quality of this product compares well with other products developed by our organization.                | 5.79 (1.08)    |
| Customer complaints have been received regarding the poor performance of this product [R].                  | 5.89 (1.31)    |
| <b>Job satisfaction</b>                                                                                     |                |
| Our work on the NPD project was stimulating                                                                 | 5.74 (0.97)    |
| The project provided us with a sense of worthwhile accomplishment.                                          | 5.33 (1.07)    |
| It provided many opportunities for personal growth and development.                                         | 5.50 (0.97)    |
| We were satisfied with the way we worked as a team.                                                         | 5.35 (1.09)    |
| <b>Firm size</b>                                                                                            | 529.1 (1177.8) |
| Number of employees in the firm.                                                                            |                |
| <b>Team size</b>                                                                                            | 7.49 (6.20)    |
| Number of people on the team who were fully involved in the project.                                        |                |

NOTE. Scale items were measured with a seven-point scale where 1 is “totally disagree” and 7 “completely agree”.

[R]: Reverted item.

**Table 4**

Zero-order correlations and discriminant validity.

|                            | CR   | AVE  | 1      | 2      | 3       | 4      | 5      | 6    | 7      | 8 |
|----------------------------|------|------|--------|--------|---------|--------|--------|------|--------|---|
| 1. Team boundary spanning  | -    | -    | -      |        |         |        |        |      |        |   |
| 2. Team boundary buffering | 0.83 | 0.61 | 0.26** | 0.78   |         |        |        |      |        |   |
| 3. Job stress              | 0.88 | 0.60 | 0.23** | 0.09   | 0.77    |        |        |      |        |   |
| 4. New product creativity  | 0.92 | 0.71 | 0.29** | 0.23** | 0.05    | 0.84   |        |      |        |   |
| 5. New product quality     | 0.73 | 0.51 | 0.05   | 0.12   | -0.31** | 0.20*  | 0.71   |      |        |   |
| 6. Job satisfaction        | 0.79 | 0.57 | 0.04   | 0.09   | -0.30** | 0.32** | 0.40** | 0.75 |        |   |
| 7. Firm size               | -    | -    | 0.13   | 0.02   | 0.04    | 0.00   | -0.06  | 0.11 | -      |   |
| 8. Team size               | -    | -    | 0.13   | -0.02  | 0.18*   | 0.01   | -0.17  | 0.04 | 0.36** | - |

Significance levels:

CR: Composite Reliability; AVE: Average Variance Extracted.

NOTE. CR and AVE of the boundary spanning dimensions are all above 0.85 and 0.50, respectively.

\* p &lt; .05.

\*\* p &lt; .01

p &lt; 0.05).

To better interpret the moderating effects, we calculated the impact of job stress on new product quality, new product creativity, and job satisfaction at high (+1 SD) and low (-1 SD) levels of boundary buffering. Regarding new product quality, findings from the analyses show that job stress has a nonsignificant effect on new product quality when boundary buffering is high and a negative and significant effect when boundary buffering is low ( $\beta = -0.60$ ,  $p < 0.01$ ) (Fig. 2). With regard to new product creativity, we found a positive and significant effect of job stress on new product creativity when boundary buffering is high ( $\beta = 0.27$ ,  $p < 0.05$ ) and a negative and significant effect when boundary buffering is low ( $\beta = -0.33$ ,  $p < 0.01$ ) (Fig. 3). Finally, while the impact of job stress on job satisfaction is not significant at high levels of boundary buffering, the effect becomes negative when boundary buffering is low ( $\beta = -0.53$ ,  $p < 0.01$ ) (Fig. 4).

## 5.2. Additional analyses

Based on H1 and H2a-c, this research implicitly proposes that job stress negatively mediates the relationships between team

**Table 5**  
Standardized parameter estimates.

|                                                               | Model 1      | Model 2     |
|---------------------------------------------------------------|--------------|-------------|
| Team boundary spanning → Job stress                           | 0.20* (H1)   | 0.20*       |
| Job stress → NP creativity                                    | −0.03 (H2a)  | −0.03       |
| Job stress → NP quality                                       | −0.35* (H2b) | −0.35*      |
| Job stress → Job satisfaction                                 | −0.33* (H2c) | −0.33*      |
| Team boundary spanning * Team boundary buffering → Job stress |              | −0.06 (H3)  |
| Job stress * Team boundary buffering → NP creativity          |              | 0.30* (H4a) |
| Job stress * Team boundary buffering → NP quality             |              | 0.25* (H4b) |
| Job stress * Team boundary buffering → Job satisfaction       |              | 0.20* (H4c) |
| Team boundary spanning → NP creativity                        | 0.26**       | 0.26**      |
| Team boundary spanning → NP quality                           | 0.17         | 0.17        |
| Team boundary spanning → Job satisfaction                     | 0.09         | 0.09        |
| Team boundary buffering → Job stress                          | 0.09         | 0.08        |
| Team boundary buffering → NP creativity                       | 0.12         | 0.19        |
| Team boundary buffering → NP quality                          | 0.04         | 0.10        |
| Team boundary buffering → Job satisfaction                    | 0.06         | 0.11        |
| R <sup>2</sup> Job stress                                     | 0.06         | 0.06        |
| R <sup>2</sup> New product creativity                         | 0.11         | 0.17        |
| R <sup>2</sup> New product quality                            | 0.14         | 0.18        |
| R <sup>2</sup> Job satisfaction                               | 0.12         | 0.14        |

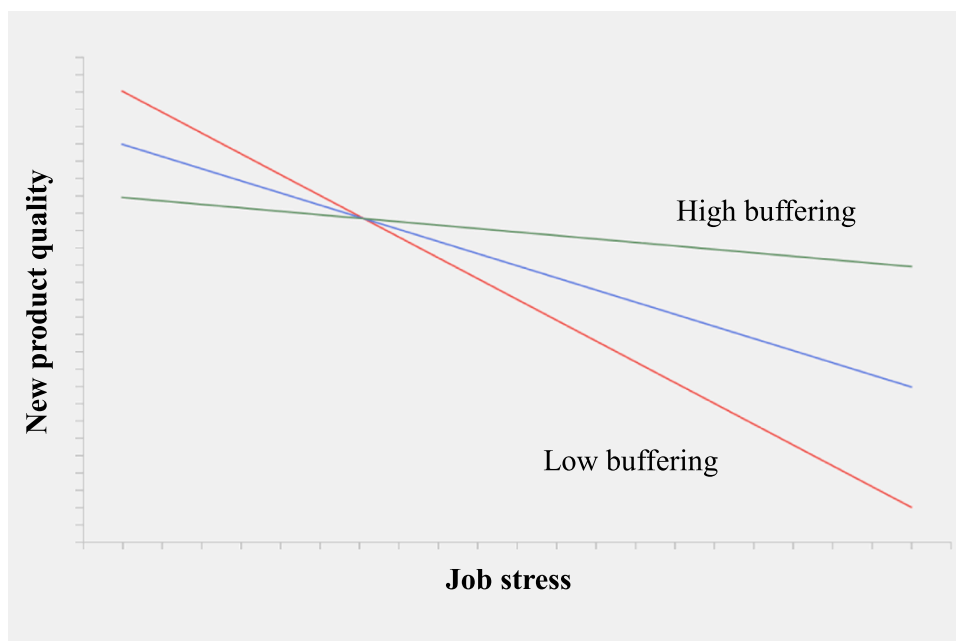
Significance levels:

Note: While we control for the effects of team size and firm size on job satisfaction, new product creativity, and new product quality, none of these effects are significant. Therefore, for brevity, these effects are not shown in the table.

\*  $p < .05$ .

\*\*  $p < .01$ .

+  $p = 0.07$  (one-tailed test).



**Fig. 2.** Moderating effect of boundary buffering on the relationship between job stress and new product quality.

boundary spanning, new product quality, new product creativity and job satisfaction. A bootstrapping procedure (5000 sub-samples) was applied to formally test the mediating effects. The bootstrapping results revealed a non-significant indirect effect of team boundary spanning on new product creativity via job stress and negative and significant indirect effects of team boundary spanning on both new product quality ( $\beta = -0.07$ ,  $p < 0.05$ ) and job satisfaction ( $\beta = -0.07$ ,  $p < 0.05$ ) through job stress.

In addition, because of the significant mediation and moderating effects of job stress and team boundary buffering respectively, we examined the possibility of moderated mediation between job stress and team boundary buffering. To formally test this moderated mediation effect, we applied the PROCESS macro proposed by Hayes (2017). This procedure allows the conditional indirect effects to

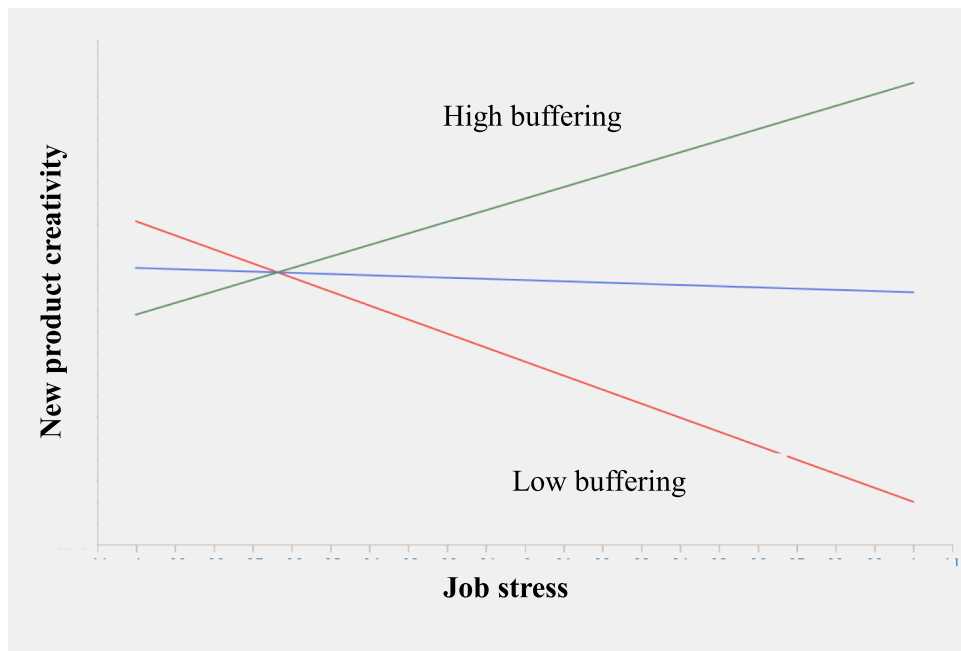


Fig. 3. Moderating effect of boundary buffering on the relationship between job stress and new product creativity.



Fig. 4. Moderating effect of boundary buffering on the relationship between job stress and job satisfaction.

**Table 6**

Indirect effects of team boundary spanning on team performance at different levels of team boundary buffering.

|                                                              | Level of boundary buffering |        |        |
|--------------------------------------------------------------|-----------------------------|--------|--------|
|                                                              | High                        | Medium | Low    |
| Team boundary spanning → job stress → New product creativity | 0.04                        | −0.01  | −0.09  |
| Team boundary spanning → job stress → New product quality    | −0.01                       | −0.07* | −0.15* |
| Team boundary spanning → job stress → Job satisfaction       | −0.02                       | −0.07* | −0.14* |

Significance levels:

\*  $p < .05$  (one-tailed test).

be estimated using bootstrap analyses (5000 samples). The PROCESS macro results showed differences in the significance of the indirect effects of team boundary spanning on new product quality and job satisfaction via job stress across high, medium and low levels of team boundary buffering (see Table 6). Specifically, we found that the negative indirect effects of team boundary spanning on both new product quality and job satisfaction become not significant at high levels of boundary buffering. However, at low levels of boundary buffering, these effects become more negative.

## 6. Discussion

Findings from this study provide answers to the pressing research questions: Does team boundary spanning impact job stress within NPD teams, and if so, how can these effects be mitigated? Regarding the first question, our findings reveal that engaging in team boundary spanning can indeed create feelings of stress. Boundary spanning activities are effortful and time consuming, and thus can absorb substantial amounts of a team's valuable resources (i.e., time, attention, energy, commitment) (Gibson and Dibble, 2013; Marrone et al., 2007; Leicht-Deobald et al., 2023). Additionally, team boundary spanning exposes team members to a broader environment of interactions, increasing the risk of receiving an overload of external requests and interferences (de Vries et al., 2022; Zhang et al., 2022). Attending to these demands can reduce the resources available for NPD team members to complete the NPD project, thereby increasing stress levels. Consequently, based on the COR theory (Hobfoll et al., 2018; Wright and Hobfoll, 2004), teams engaged in boundary spanning are likely to experience elevated levels of stress.

Moreover, the results show that job stress can have a negative impact on new product quality and job satisfaction. Job stress can be overpowering, negatively affecting individuals' competence in aspects of information processing (Nygaard and Dahlstrom, 2002), communication and coordination (Barczak and Wilemon, 2003), as well as their desire to engage in learning behaviors (Savelsbergh et al., 2012), all of which are critical for maintaining high product quality. Similarly, when under stress, employees' engagement and their feelings of accomplishment and growth can be diminished, with a subsequent negative impact on team satisfaction (Abubakar et al., 2022).

Unlike with new product quality and job satisfaction, we did not find a significant effect of job stress on new product creativity. This can be contextualized within the framework of challenge-hindrance stressors, which highlights the dual nature of stress effects (LePine et al., 2005; Podsakoff et al., 2007). While stress is often viewed negatively, the literature suggests that certain stressors, perceived as challenges, can actually foster creativity by stimulating problem-solving and idea generation. Challenge stressors, though demanding, are associated with personal growth and achievement, and are believed to stimulate creative efforts by pushing individuals to engage in problem-solving and idea generation, thereby potentially enhancing creativity (Cavanaugh et al., 2000). However, in our study, we anticipated that stress would primarily act as a hindrance, obstructing creative processes rather than fostering them. Hindrance stressors are perceived as obstacles that overwhelm individuals and detract from their ability to engage in creative thinking (Abbas and Raja, 2019; Cavanaugh et al., 2000; Gutnick et al., 2012). Given this perspective, the absence of a significant negative effect of job stress on new product creativity suggests the need for further investigation into how NPD teams perceive the stress generated by boundary spanning activities.

Regarding the moderating effect of team buffering, our findings did not reveal a significant moderating effect of team buffering on the relationship between team boundary spanning and job stress. The absence of a significant moderating effect may be attributed to the timing and anticipatory actions of NPD teams in response to stress. Typically, at the outset of a NPD project, teams may not fully recognize or predict the stressful outcomes that could arise from boundary-spanning activities. Their initial focus is more likely to be on the opportunities and acquisition of new resources these activities offer (Leicht-Deobald et al., 2023) than on mitigating potential stress. As the project advances and the stress caused by boundary spanning becomes more evident, teams might then start to implement boundary buffering strategies (de Vries et al., 2022). This reactive nature of boundary buffering could explain why we did not observe a significant moderating effect on the relationship between team boundary spanning and job stress; buffering strategies may not be in place early enough to effectively mitigate the impact of boundary spanning on job stress. The tendency to delay buffering may also be influenced by the perception that boundary buffering is secondary and less crucial compared to boundary spanning. Leicht-Deobald et al.'s (2023) meta-analysis on team boundary management highlights that significantly more studies have focused on boundary spanning than on boundary buffering, and there is general consensus on the positive impact of the former on team performance. As a result, NPD teams may prioritize boundary spanning over buffering activities, delaying the implementation of buffering strategies until stress has already escalated. This delay is likely to reduce the effectiveness of buffering in mitigating job stress caused by boundary spanning.

However, our results show that team boundary buffering moderates the relationships between job stress and NPD performance and job stress and team satisfaction. In this respect, the study's results indicate that high levels of boundary buffering can eliminate the negative effects that job stress exerts on both new product quality and job satisfaction. By providing NPD teams with valuable resources (e.g., psychological safety, emotional energy, time, cognitive attention, enthusiasm) (Faraj and Yan, 2009; Leicht-Deobald et al., 2022), team boundary buffering helps NPD teams replace or reinforce the resources lacking as a consequence of experiencing stress, thus assisting NPD teams to adapt and better cope with felt stress. Together, these results suggest that while team boundary buffering may not be an effective proactive stress intervention, it can work well as a reactive intervention to the stress created by team boundary spanning.

Concerning new product creativity, we observed a non-monotonic relationship between boundary buffering, job stress, and new product creativity. Specifically, job stress positively influences new product creativity when boundary buffering is high, but it has a negative effect when boundary buffering is low. These findings align with the theoretical work conducted by Gutnick et al. (2012) on the impact of stressors on creative performance. According to Gutnick et al., individuals' perception of stressors as either challenges or

hindrances depends on their expectations regarding how manageable a situation is. If employees believe they have sufficient resources to cope with the demands of a highly stressful situation, their perception of manageability is likely to be enhanced. Following this idea, we posit that under conditions of high boundary buffering, when sufficient resources are expected to be available to help teams address the demands of a stressful situation, team members are more likely to perceive stress as a challenge, and thus benefit from the positive effect that challenge appraisals have on new product creativity. In contrast, under conditions of low boundary buffering where teams may not have enough resources to manage the demands of a highly stressful situation, job stress is likely to be perceived as a hindrance, thus negatively impacting new product creativity.

## 7. Theoretical and managerial implications

This study makes several key theoretical contributions to the literature on team boundary management and NPD. Drawing on Colquitt and Zapata-Phelan's (2007) taxonomy of theoretical contributions, we position our work as both building and testing theory in significant ways.

First, this research expands the literature on team boundary spanning by focusing on its potential negative consequences—an area that has received limited attention. To date, existing studies have predominantly emphasized the positive aspects of team boundary spanning, such as improved access to resources and enhanced team performance (Dey and Ganesh, 2017; Marrone, 2010), with little attention paid to its negative effects. While some research has explored the negative consequences of boundary spanning, these studies have primarily focused on the individual level, where both boundary spanning and its outcomes (e.g., role overload, information overload, and role stress) are measured at the individual level (Marrone et al., 2007; Wang and Zhang, 2016; Zhang et al., 2022; Zhu et al., 2020). One notable exception to this individual-level focus is the work of Gibson and Dibble (2013), which investigated an inverted U-shaped relationship between boundary spanning and team effectiveness—but their study was conducted in the context of filmmaking teams rather than NPD. In contrast, our study measures both boundary spanning and its negative effects at the team level, offering a novel perspective by examining how team boundary spanning leads to job stress, which subsequently impacts job satisfaction and NPD performance. By doing so, our findings contribute to theory building by expanding the understanding of boundary spanning to include its potential costs in NPD team settings, challenging the predominant view of it as a largely beneficial activity.

Second, this study validates and extends the applicability of Conservation of Resources (COR) theory in NPD team contexts. COR theory (Hobfoll, 1989; Hobfoll et al., 2018) posits that individuals strive to protect and build resources, and stress arises when there is a perceived or actual loss of these resources. By hypothesizing and empirically demonstrating how boundary spanning depletes team resources and triggers stress, our study contributes to theory testing, as described by Colquitt and Zapata-Phelan (2007).

Third, this research is among the first to explore the interaction between team boundary spanning and boundary buffering as complementary activities. Prior studies have examined these boundary management strategies in isolation (Faraj and Yan, 2009; Leicht-Deobald et al., 2022), with little attention paid to how they interact to influence team performance. Our findings reveal that while boundary buffering is ineffective as a proactive measure, it serves as a critical reactive intervention when boundary spanning has led to elevated stress levels. This underscores the importance of adopting a holistic approach to boundary management (Kislov, 2018), which advances boundary management theory by demonstrating that boundary buffering and spanning can be complementary rather than conflicting strategies (Dey and Ganesh, 2017; Drach-Zahavy and Somech, 2010). By showing that these two activities can be strategically aligned, our study builds new theoretical insights into how teams can effectively manage the pressures associated with boundary spanning to maintain satisfaction and performance.

Our research also carries significant managerial implications. Firstly, the findings underscore the importance for NPD managers to recognize and consider the costs associated with team boundary spanning. While engaging in boundary spanning activities brings benefits to NPD teams – for example, findings from this study highlight a positive effect on new product creativity –, managers should also be mindful of the negative consequences. Specifically, our study reveals that team boundary spanning can lead to feelings of stress among team members, which in turn have a detrimental effect on new product quality and job satisfaction. Consequently, it is critical for managers to not only promote boundary spanning for its benefits but also implement strategies to counteract its potential drawbacks.

In this study, we have identified that boundary buffering activities can significantly alleviate the adverse effects of stress stemming from boundary spanning. Given this finding, it is imperative for NPD managers to foster robust buffering competencies within their teams. To achieve this, managers should consider developing comprehensive training programs tailored to equip team members with the necessary skills for effective boundary buffering (Cross et al., 2000). Such training could encompass a variety of interactive and practical exercises, including the use of real-life case studies and simulated scenarios. These training sessions could focus on teaching NPD teams tactical disengagement strategies, enabling them to decline non-essential interactions thoughtfully, without jeopardizing relationships with important stakeholders. Additionally, these exercises could guide team members on how to effectively prioritize tasks and requests to align with the overarching team goals and project requirements.

Implementing roles such as gatekeepers within teams can further enhance the management of incoming information and demands, ensuring that only pertinent and crucial information reaches team members (Yu et al., 2022). Managers can also formalize policies that outline clear procedures for handling external requests, establishing guidelines that prioritize these requests based on their urgency and relevance to project objectives. Moreover, the strategic implementation of digital tools that allow teams to tag and categorize incoming information and requests can significantly streamline the management of external interactions, reducing the cognitive load on team members (Zhang et al., 2022).

Furthermore, the adoption of designated 'quiet hours', during which external communications are minimized, can be a practical approach to reduce disruptions and maintain focus (König et al., 2013). This practice helps create a controlled environment that

supports deep work and minimizes stress, thereby enhancing overall team productivity and satisfaction. Collectively, these buffering strategies can mitigate the detrimental impact of job stress induced by boundary spanning on job satisfaction and new product quality.

## 8. Limitations and future research lines

This study has certain limitations. First, the team leaders of the selected NPD projects provided the data for this study on the basis that team leaders would have first-hand knowledge of the team's boundary management practices as well as the levels of job stress and NPD team performance. However, it is possible that the perspective of the leader could differ from that of team members and thus, we suggest that future studies collect data from multiple members within each team. Secondly, we used a retrospective methodology which is susceptible to recall bias (Brockman and Morgan, 2003). Furthermore, while our research explores causal relationships, the cross-sectional design of the survey study restricts our ability to test causality. To examine the causal relationships proposed in this study, longitudinal designs are required. Finally, our study is focused on NPD projects developed in high-technology and medium-high technology industries within Spain, which may limit the generalizability of our results. Regarding the geographical location, Spain's parameters are comparable to the average of OECD countries in terms of innovation (with Spain's Global Innovation Index at 45.9 in 2023 compared to the OECD's 48.6) and economic development (with Spain's GDP growth at 2.5 % in 2023 versus 1.6 % for OECD members). This comparison suggests that the results obtained might be similar to those in other OECD countries. However, extending the study to include low-technology industries could be valuable, as job stress may be lower in these sectors where NPD teams typically manage fewer projects. Furthermore, considering Lan et al. (2022) findings that the impact of boundary spanning on individuals' job satisfaction, performance, innovation, and role stress is more pronounced in collectivist and high-power distance cultures than in individualist and low power distance contexts, conducting studies in countries with different levels of power distance and collectivism than Spain would be beneficial to confirm our findings.

In addition to addressing these limitations, we also identify other future research lines. First, buffering activities are only one of two dimensions of team boundary strengthening –i.e., which refers to the actions taken by the team to defend the team's activity borders. In addition to boundary buffering, Dey and Ganesh (2017) identify a second strengthening dimension: reinforcement. This relates to the activities through which the group boosts commitment, embracing a sense of belonging amongst those within the team (Cross et al., 2000). In this respect, it might be interesting to examine the interplay between boundary buffering and reinforcement. In particular, as shown in this study, merely carrying out buffering activities does not seem enough to minimize the positive effect of team boundary spanning on job stress. Thus, it may be possible for buffering activities to become more effective if accompanied by reinforcement activities (Faraj and Yan, 2009). Secondly, Wang et al. (2019) distinguish between transactional boundary spanning (routine tasks) and learning boundary spanning (non-routine tasks). Thus, future studies could differentiate between these two types of boundary spanning activities and analyse how each of them affects job stress. Finally, because both boundary spanning and boundary buffering activities change over the course of the project (Drach-Zahavy and Somech, 2010; Leitch-Deobald et al., 2022; Day and Ganesh, 2017; Wu et al., 2020), it may be interesting to study the interplay between team boundary spanning and boundary buffering throughout the different stages of a NPD project. This research line might help us to deepen our knowledge about when it is appropriate to commence boundary buffering activities in an NPD project so as to optimize its results.

## CRedit authorship contribution statement

**Rodríguez-Escudero Ana Isabel:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Carbonell Pilar:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Temprano García Victor:** Writing – review & editing, Writing – original draft. **Pérez-Fernández Hector:** Writing – review & editing, Writing – original draft.

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## Data availability

The authors do not have permission to share data.

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